

HFT / Order Flow Project Consulting Review

Senior Quant Research Perspective (35+ Years Equivalent Review)

Overall Assessment

You are going in the right direction. The most impressive aspect is not the RF MCC of 0.31, but the engineering discipline: finding and fixing contamination, rebuilding the pipeline, documenting root cause analysis, running leakage tests, rejecting data snooping, and using stock-level cross validation.

The biggest risk is not the model. The biggest risk is optimizing a dataset with only one trading day of data. The current model may be learning stock characteristics rather than predictive order flow dynamics.

Most Important Immediate Experiment

Neutralize stock-level effects by z-scoring features within each stock or subtracting per-stock means. Retrain the model. If MCC collapses, the model was learning stock identity. If performance remains largely intact, you likely have genuine temporal signal.

Label Design Critique

A 60-minute 5% movement target is more suitable for swing trading than microstructure prediction. HFT firms typically predict next trade direction, short-horizon mid-price movement, expected return, and then determine tradability after costs.

Add continuous return targets (5m, 15m, 30m, 60m), cross-sectional ranking labels, and triple-barrier labels in addition to the current classification setup.

Missing Features

Priority additions: Microprice, true Order Flow Imbalance (OFI), queue imbalance at multiple depths, order book slope, queue refill speed across multiple horizons, trade size bucket distributions, and volatility regime indicators.

Trade Reconstruction Validation

Do not proceed further without validating inferred trades against actual tick data. Compare inferred versus real buy/sell volume per minute using correlation, MAE, MAPE, and R^2 . If correlation is below approximately 0.8–0.9, the inferred trade stream may be introducing substantial noise.

RF MCC = 0.31

The result is believable but not yet trustworthy. One-day datasets frequently produce optimistic estimates. It would not be surprising for MCC to decline materially after introducing multiple weeks of data.

Data Requirements

The 3,000 stock-day target is reasonable. A stronger target is 10,000+ stock-days, for example 500 stocks across 20 trading days. This is where real-world robustness testing begins.

Recommended Model Progression

1. Logistic Regression (keep forever as baseline) 2. LightGBM (likely strongest practical model) 3. CatBoost 4. Small MLP after large-scale data collection 5. Transformers only much later

Observability Framework

Track data quality, feature drift, label balance, MCC, precision, recall, calibration, simulated PnL, Sharpe ratio, hit rate, turnover, and expected versus realized alpha on a daily basis.

How Professional Firms Operate

Approximate effort allocation: 60% Data, 25% Validation, 10% Infrastructure, 5% Modeling. Most performance improvements come from better data and validation rather than more sophisticated models.

Recommended Next Six Steps

1. Validate inferred trades against real ticks. 2. Neutralize stock-level effects and rerun experiments. 3. Acquire 20–30 trading days of data. 4. Add microprice, OFI, and queue imbalance features. 5. Add continuous-return targets alongside classification labels. 6. Benchmark Logistic Regression versus LightGBM on multi-day data.